EE 5239 Nonlinear Optimization

Lecture 13c: Sparse and Low Rank Optimization

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Outline

Announcement

- HW 5 due tonight (Nov 27th).
- HW 6 will be released today and due Dec 16th.
- Due date for the project report (and codes) will be the last day of the final exam (Dec 19.)

Last Time

- Mainly focused on sparse modeling
- Motivation behind sparse models

$$\|\boldsymbol{x}\|_1 = \sum_{k=1}^{K} |x_k|$$

- What types of regularization induces which types of sparsity
  1. $R(\boldsymbol{x}) = \|\boldsymbol{x}\|_1$, tend to produce $\boldsymbol{x}$ vector with a few nonzeros
  2. $R(\boldsymbol{x}) = \|\boldsymbol{x}\|_2$, tend to produce $\boldsymbol{x}$ vector with all zeros (only kink $\boldsymbol{x} = 0$)
  3. $R(\boldsymbol{x}) = \sum_{I=1}^{G} \|\boldsymbol{x}_{[I]}\|_2$, where $\boldsymbol{x}_{[I]}$ is a group of variables; tend to produce $\boldsymbol{x}$ vector with a few groups of nonzeros
  4. $R(\boldsymbol{x}) = \sum_{k=1}^{K-1} |x_k - x_{k+1}|$, tend to produce $\boldsymbol{x}$ vector with a few nonzeros adjacent differences

Last Time

- The Proximal Gradient Descent Method

$$\begin{aligned}\boldsymbol{x}^{r+1} &= \arg\min_{\boldsymbol{x}} \langle \nabla g(\boldsymbol{x}^r), \boldsymbol{x} - \boldsymbol{x}^r \rangle + \frac{L}{2}\|\boldsymbol{x} - \boldsymbol{x}^r\|^2 + \lambda R(\boldsymbol{x}) \\ &= \arg\min_{\boldsymbol{x}} \frac{L}{2}\|\boldsymbol{x} - (\boldsymbol{x}^r - \frac{1}{L}\nabla g(\boldsymbol{x}^r))\|^2 + \lambda R(\boldsymbol{x}) \\ &:= \text{prox}_R^{\lambda/L}\left(\boldsymbol{x}^r - \frac{1}{L}\nabla g(\boldsymbol{x}^r)\right)\end{aligned}$$

- For LASSO and Group LASSO, we see how the proximity operator can be computed in closed form
- Coordinate Descent is also a popular algorithm

This Time

- We discuss structured optimization for problems involving matrices
- How (and why) to encourage sparsity for matrices?
- How (and why) to encourage low-rankness in matrices?
- Good algorithms and applications

Application: Weight Decay

$$\min_{W} \quad L_{\text{new}}(W) = L_{\text{original}}(W) + \lambda \|W\|_{\text{fro}}^2$$

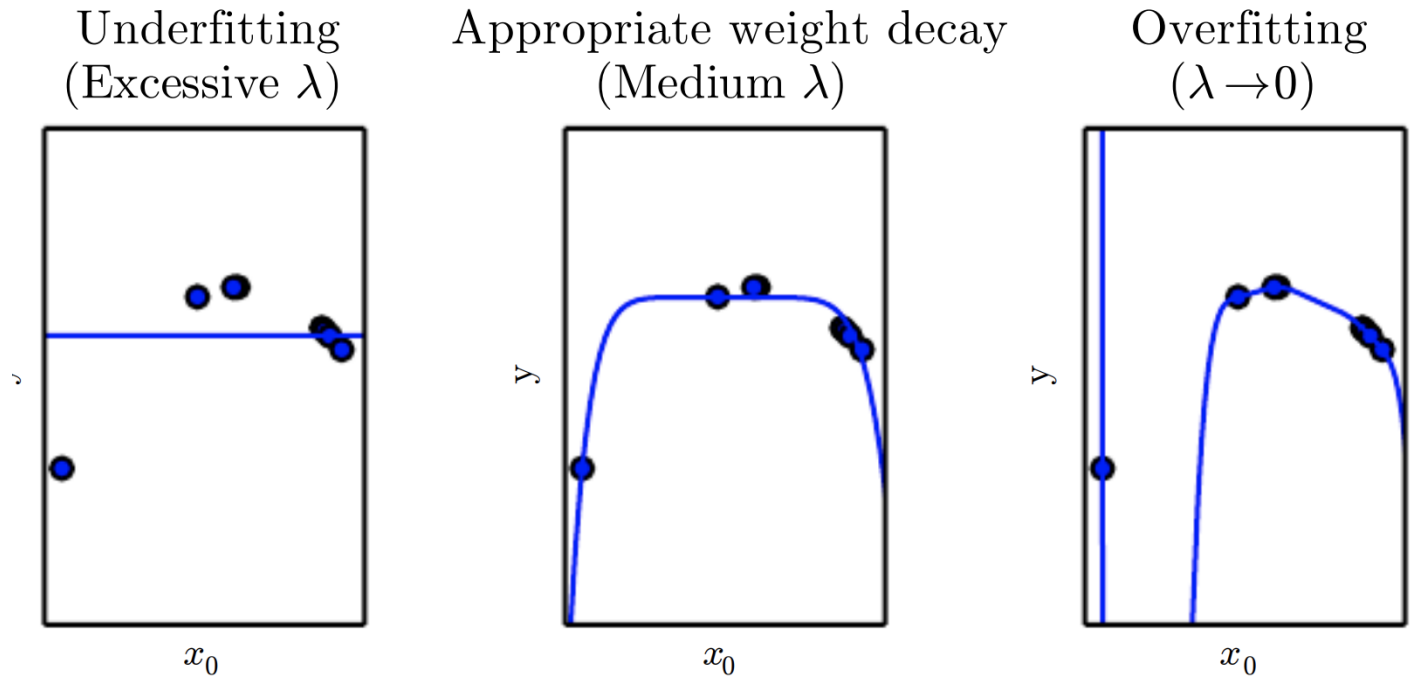
- W is the weight matrix
- Proximal operator of $R(\cdot) = \lambda \|\cdot\|_{\text{fro}}^2$ is

$$\text{prox}_R^{\beta}(X) = \frac{1}{1 + 2\beta\lambda} X$$

- Shrink the weight uniformly by a factor $1 + 2\beta$
- "Ridge regression"

Application: Weight Decay

- What do we gain by shrinking the weights?
- Prevent overfitting



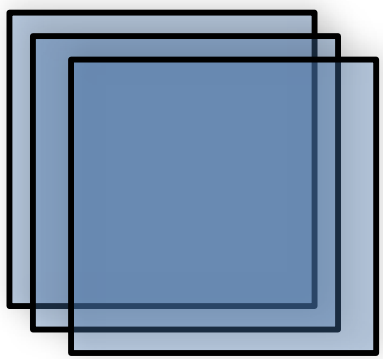
- Right now ℓ_1 regularization is not popular in deep learning. Why?

Application: Anomaly Detection

- **Objective.** Find unusual and rare events from a large amount of data
- Lots of applications, data analysis, imaging, cybersecurity
- **Ways.** Utilizing the sparsity of the "anomaly"

Application: Anomaly Detection

- An application on foreground separation from noisy surveillance video
- Suppose we have a sequence of video frames $\{\mathbf{M}^i\}_{i=1}^n$
- First we do preprocessing:



Application: Anomaly Detection

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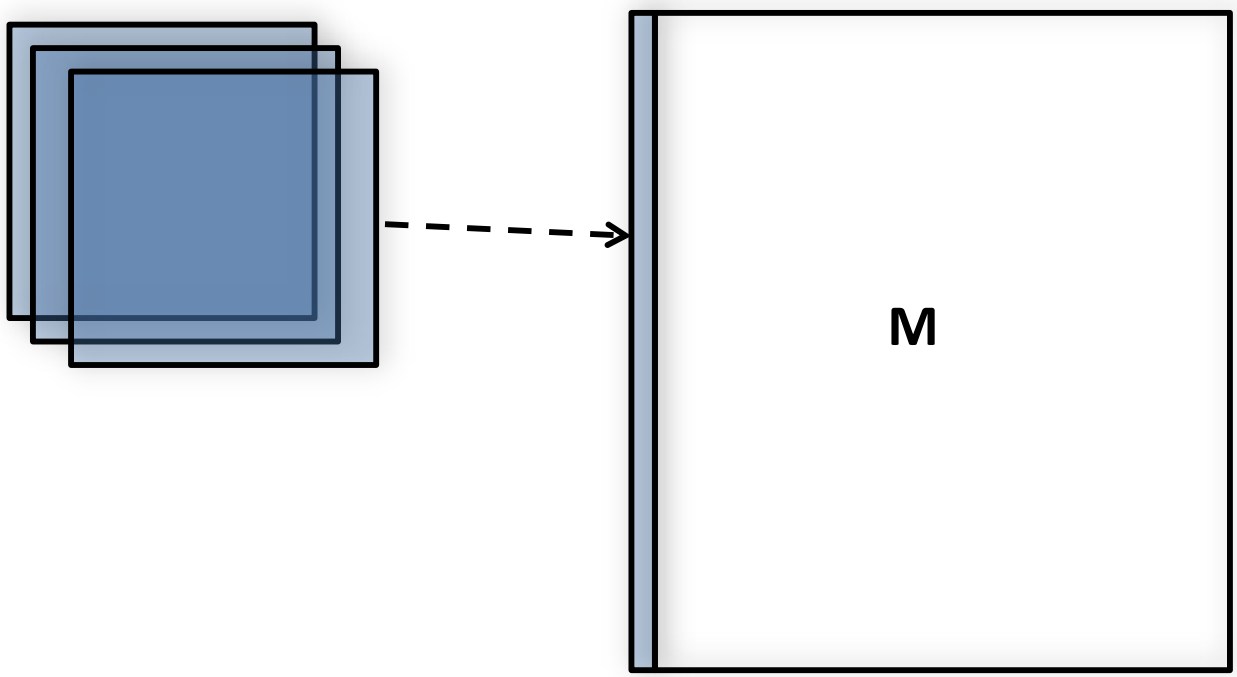

Application: Anomaly Detection

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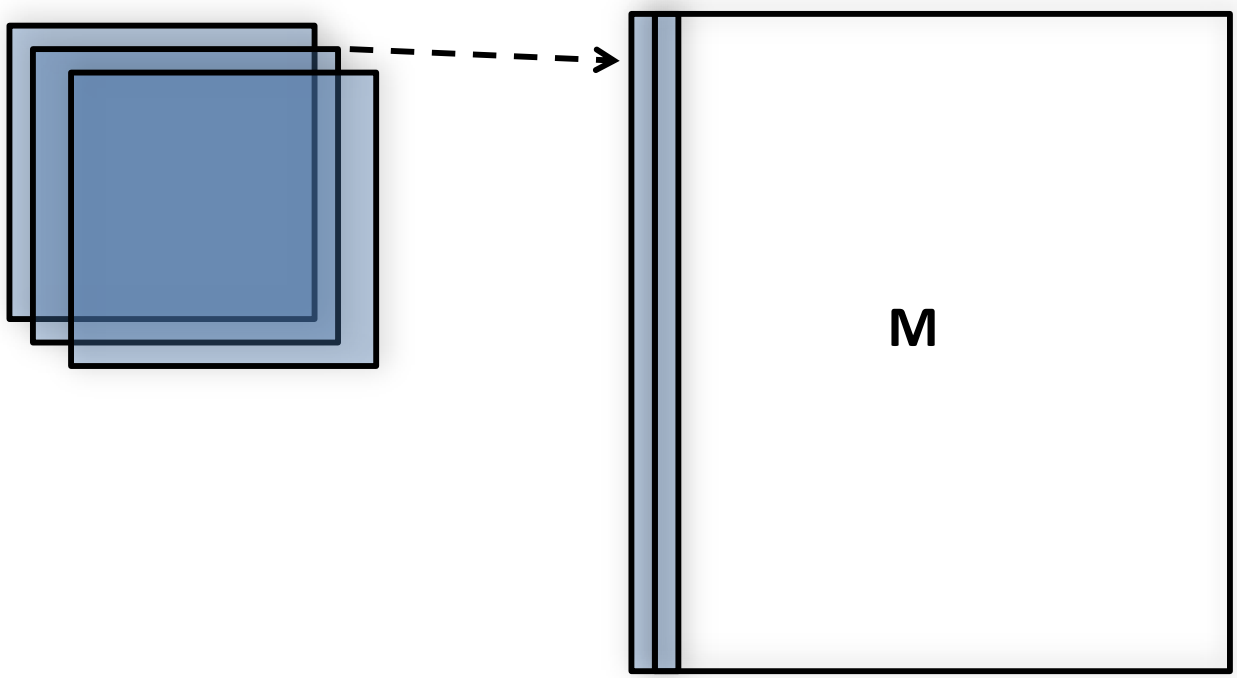


Application: Anomaly Detection

- We make each column of $\mathbf{M}$ becomes one frame of video
- **Objective**: Decompose it into three parts:
  1. $\mathbf{L}$ is the background (low rank)
  2. $\mathbf{S}$ is the moving foreground (sparse)
  3. $\mathbf{Z}$ is the noise (small magnitude)
- Why low rank matrix represents background? Changes slowly; consider a matrix with identical columns, the rank is 1
- **Note**: No "label", unsupervised learning!
- **Main Idea**: Promote these properties by penalizing using different costs: $(\min\ \mathrm{Rank}(\mathbf{L}) + \mathrm{Density}(\mathbf{S}) + \mathrm{Magnitude}(\mathbf{Z}))$

Application: Anomaly Detection

$$\begin{aligned} \min \quad & \|\mathbf{L}\|_* + \gamma_1 \|\mathbf{S}\|_1 + \gamma_2 \|\mathbf{Z}\|_F^2 \\ \text{subject to} \quad & \mathbf{L} + \mathbf{S} + \mathbf{Z} = \mathbf{M} \end{aligned}$$

- 250×135 pixels per frame, 50 frames, over 1 million variables!!
- Similar idea applies to
  1. Analyzing power network security (intrusion detection)
  2. Data cleansing (getting rid of the data outliers)
  3. Network data flow with intrusion detection

Application: Anomaly Detection

$$\begin{aligned} \min_{\mathbf{L},\mathbf{S},\mathbf{M}} \quad & \|\mathbf{L}\|_* + \gamma_1 \|\mathbf{S}\|_1 + \gamma_2 \|\mathbf{Z}\|_F^2 \\ \text{subject to} \quad & \mathbf{L} + \mathbf{S} + \mathbf{Z} = \mathbf{M} \end{aligned}$$

$$\min_{\mathbf{L},\mathbf{S}} \quad \|\mathbf{L}\|_* + \gamma_1 \|\mathbf{S}\|_1 + \gamma_2 \|\mathbf{M} - \mathbf{L} - \mathbf{S}\|_F^2$$

1. How to solve it?
2. Fix $\mathbf{L}$ update $\mathbf{S}$; fix $\mathbf{S}$ update $\mathbf{L}$

Application: Anomaly Detection

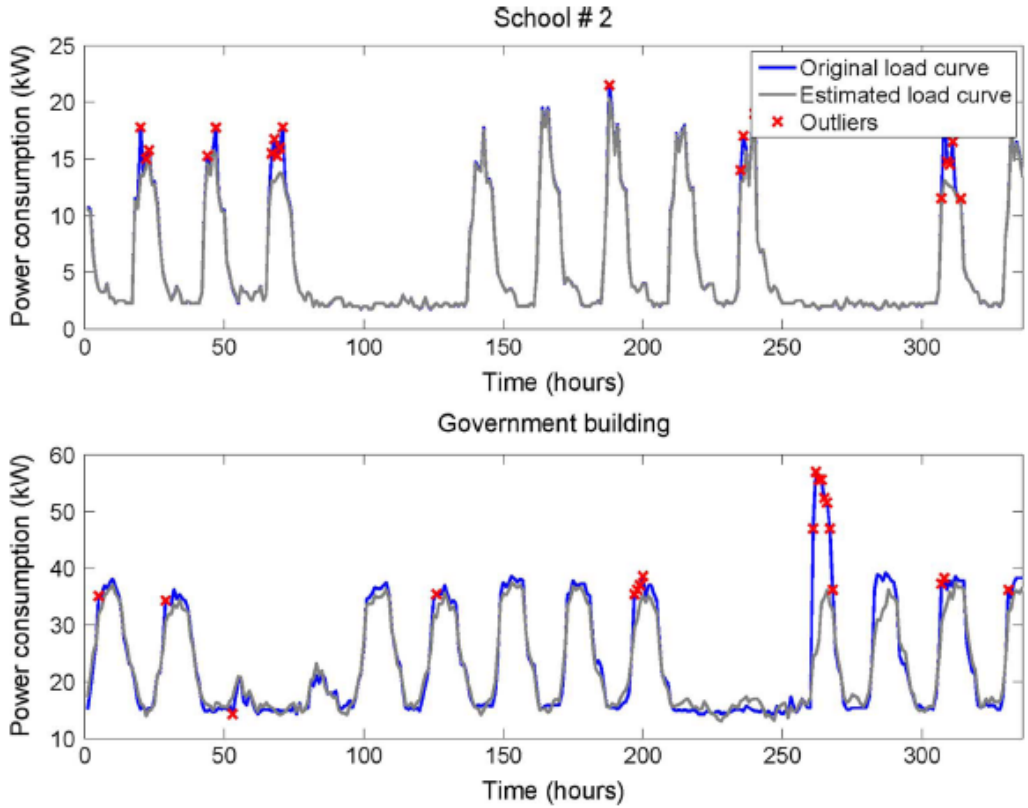


Figure 0.1: An Example in Power Data Cleansing and Imputation [Mateos and Giannakis 13]

Applications: Collaborative Filtering

1. Netflix predicts other "Movies You will love" based on past numeric ratings (1-5 stars)
2. Recommendations drive 60% of Netflix's DVD rentals

Figure 0.2: [Mackey 2009]

Applications: Collaborative Filtering

1. Netflix Prize: Beat Netflix recommender System, using Netflix Data, Win $ million
2. Data: 480,000 users, 18,000 movies, 100 mil observed ratings = only 1.1% of ratings observed

Figure 0.3: [Mackey 2009]

Example: Collaborative Filtering

1. **Question**: Will user i like movie j?
2. **Collaborative Filtering**: Filtering information using techniques involving collaboration among multiple agents, view points, data sources.
3. **Matrix Completion**: Complete the matrix based on partially filled information

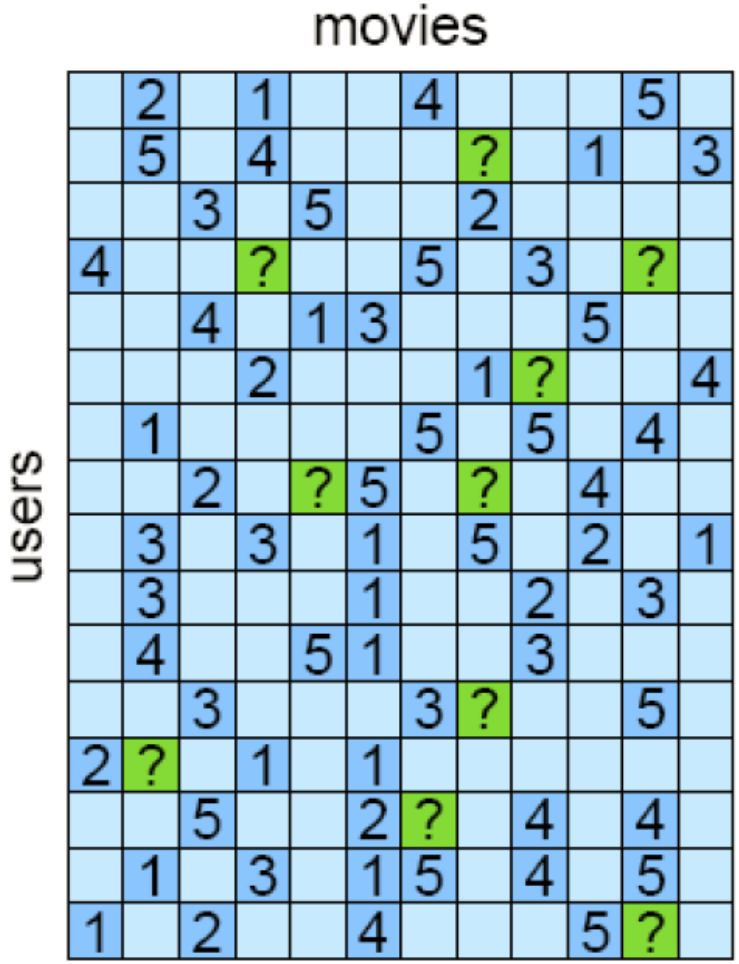


Figure 0.4: [Scheinberg 2012]

Example: Collaborative Filtering

- Intuitively, collaborative filtering is to find a latent space

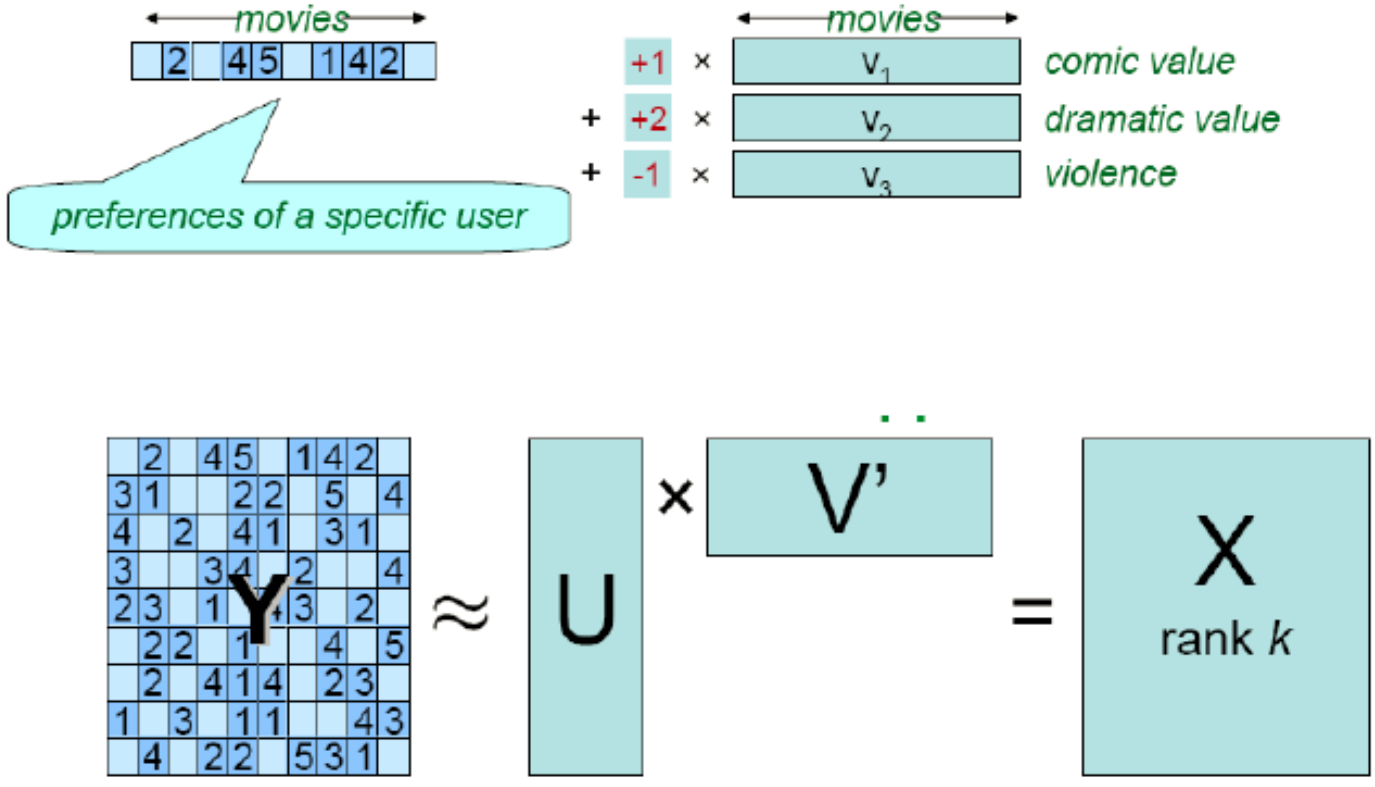


Figure 0.5: Formulation of Collaborative Filtering [Scheinberg 2012]

Example: Collaborative Filtering

- Suppose $\mathbf{M}$ is the observation, where the indices in the set $\mathcal{I}$ are those observed data points

$$\mathcal{I} := \{(i,j) \mid M_{i,j} \text{ is observed}\}$$

- We want to find a full matrix $\mathbf{X}$

Example: Collaborative Filtering

- Suppose that there are K hidden topics/factors representing each movie
- That is, a give movie i is represented by a vector

$$\mathbf{v}_i = [v_{i,1}; v_{i,2}; \ldots; v_{i,K}]$$

  1. an index k represents a movie category
  2. each $v_{i,k}$ represents the strength of movie i in category k

- A user j's rating for movie i is modeled as

$$\mathbf{X}_{ij} = \mathbf{u}_i^T \mathbf{v}_j$$

  where $\mathbf{u}_i = [u_{i,1}; \cdots, ; u_{i,K}]$ is a vector representing how much user i likes each category

Example: Collaborative Filtering

- **Key Observation.** If we can learn the users' preferences, as well as the content of each movie, then we know $\mathbf{X}_{ij}$!
- We can formulate the Collaborative Filtering problem as

$$\text{Decompose } \mathbf{X} \text{ into } \mathbf{U} \times \mathbf{V}^T$$

where the following requirements are met

1. $\mathbf{X} \approx \mathbf{M}$
2. $\mathbf{X}$ has low rank
3. The original observations are all consistent

Example: Collaborative Filtering

- First suppose we know that the rank of X is K
- We can formulate the problem as a matrix factorization problem

$$\min_{\mathbf{U}\in\mathbb{R}^{m\times K}\,\mathbf{V}\in\mathbb{R}^{n\times K}} \quad \sum_{(i,j)\in\mathcal{I}} \|\mathbf{U}_i\mathbf{V}_j^T - \mathbf{M}_{ij}\|^2$$

- Nonconvex formulation
- Algorithm: Alternating Least Squares (ALS)
  1. Fix $\mathbf{V}$, optimize $\mathbf{U}$, least square
  2. Fix $\mathbf{U}$, optimize $\mathbf{V}$, least square
- Basically a coordinate descent algorithm

Example: Collaborative Filtering

- What if we don't know the rank (or the order of the hidden factors?)
- More formally, we can write down the following optimization problem

$$\begin{aligned} \min_{X \in \mathbb{R}^{m \times n}} \quad & \mathrm{Rank}(\mathbf{X}) \\ \text{s.t.} \quad & X_{ij} = M_{ij},\ (i,j) \in \mathcal{I} \end{aligned}$$

- **Question**: What is the rank of matrix $\mathbf{X}$? Easy to optimize?

Outline

Matrices

1. Let $\mathbf{A}$ be a $m \times n$ matrix.
2. Rank of $\mathbf{A}$ $\mathrm{Rank}(\mathbf{A})$. Full rank matrix $\mathbf{A}$: $\mathrm{Rank}(\mathbf{A}) = \min\{m, n\}$.
3. (Complex) Eigenvalue λ: $\mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ for some $\boldsymbol{x} \neq 0$.
4. Spectral radius: $\rho(\mathbf{A}) = \max_i\{|\lambda_i|\}$, λ_i is an eigenvalue of $\mathbf{A}$.
5. Eigen-decomposition of a symmetric matrix: $\mathbf{A} = \mathbf{P}'\mathbf{\Lambda}\mathbf{P}$; where $\mathbf{P}$ is orthonormal, $\mathbf{\Lambda}$ is diagonal and real.
6. Trace: $\mathrm{Tr}[\mathbf{A}] = \sum_{i=1}^n \mathbf{A}_{ii} = \sum_{i=1}^n \lambda_i$

Matrices

1. Singular value decomposition (SVD) (suppose $m \geq n$):

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}' \in \mathbb{R}^{m\times n},$$

where $\mathbf{\Sigma} \in \mathbb{R}^{m\times n}$ rectangular diagonal matrix with diagonals $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$; $\mathbf{U} \in \mathbb{R}^{m\times m}$, $\mathbf{V} \in \mathbb{R}^{n\times n}$ orthonormal

2. Let us use $\sigma(\mathbf{A})$ to denote the vector of singular values

3. What is the rank of $\mathbf{A}$?

$$\mathrm{Rank}(\mathbf{A}) = \text{number of nonzero } \sigma_i\text{'s} = \|\sigma(\mathbf{A})\|_0$$

4. Trace norm/nuclear norm of a matrix $\mathbf{A}$ is

$$\|\mathbf{A}\|_* = \|\sigma(\mathbf{A})\|_1 = \sum_{i=1}^{n} \sigma_i = \mathrm{Tr}\left(\sqrt{\mathbf{A}^T \times \mathbf{A}}\right)$$

Matrices and Norms

1. Norms:

$$\text{Frobenious Norm} : \|\mathbf{A}\|_F = \left(\sum_{i,j} |A_{ij}|^2\right)^{1/2} = \left(\sum_i \sigma_i^2\right)^{1/2} \tag{0.1}$$

$$\text{Nuclear Norm} : \|\mathbf{A}\|_* = \sum_i \sigma_i \tag{0.2}$$

$$\text{Matrix 2-norm} : \|\mathbf{A}\|_2 = \sup_{\boldsymbol{x}\neq 0} \frac{\|\mathbf{A}\boldsymbol{x}\|}{\|\boldsymbol{x}\|} = \max_i \sigma_i \tag{0.3}$$

2. Corresponds to vector 2-norm of singular values ($\|\cdot\|_2$), vector 1-norm of singular values ($\|\cdot\|_1$), vector maximum norm of singular values ($\|\cdot\|_\infty$)
3. Useful inequalities:

$$\|\mathbf{A}\boldsymbol{x}\| \leq \|\mathbf{A}\|_2\|\boldsymbol{x}\|_2, \ \ \|\mathbf{A}\|_* \geq \|\mathbf{A}\|_F \geq \|\mathbf{A}\|_2$$

Back to Collaborative Filtering

- Recall that the collaborative filtering problem has been formulated as

$$\begin{aligned} \min_{X \in \mathbb{R}^{m \times n}} \quad & \mathrm{Rank}(\mathbf{X}) \\ \text{s.t.} \quad & X_{ij} = M_{ij},\ (i,j) \in \mathcal{I} \end{aligned}$$

- This is difficult problem (NP-hard, as $\mathrm{Rank}(\mathbf{X}) = \|\sigma(\mathbf{X})\|_0$)
- **Practical Solution**: Use $\|\sigma(\mathbf{X})\|_1 = \|\mathbf{X}\|_*$ instead!

$$\begin{aligned} \min_{X \in \mathbb{R}^{m \times n}} \quad & \|\mathbf{X}\|_* \\ \text{s.t.} \quad & X_{ij} = M_{ij},\ (i,j) \in \mathcal{I} \end{aligned}$$

Different Formulations

- The collaborative filtering (or the matrix completion problem) has several different formulations
- We have seen one formulation, which requires the observations M_{ij} to be exactly matched
- Other formulations includes

$$\begin{aligned} \min_{X \in \mathbb{R}^{m \times n}} \quad & \|\mathbf{X}\|_* \\ \text{s.t.} \quad & \|\mathbf{X}_{\mathcal{I}} - \mathbf{M}_{\mathcal{I}}\|^2 \le \epsilon \end{aligned} \quad \leftrightarrow \quad \min_{X \in \mathbb{R}^{m \times n}} \quad \|\mathbf{X}\|_* + \lambda \|\mathbf{X}_{\mathcal{I}} - \mathbf{M}_{\mathcal{I}}\|^2$$

$$\begin{aligned} \min_{X \in \mathbb{R}^{m \times n}} \quad & \frac{1}{2}\|\mathbf{X}_{\mathcal{I}} - \mathbf{M}_{\mathcal{I}}\|^2 \\ \text{s.t.} \quad & \|\mathbf{X}\|_* \le R \end{aligned} \quad \leftrightarrow \quad \min_{X \in \mathbb{R}^{m \times n}} \quad \lambda \|\mathbf{X}\|_* + \frac{1}{2}\|\mathbf{X}_{\mathcal{I}} - \mathbf{M}_{\mathcal{I}}\|^2$$

The Proximity Operator For Nuclear Norm

- Consider the following nuclear norm regularized problem

$$\min_{\mathbf{X}} \quad g(\mathbf{X}) + \underbrace{\lambda\|\mathbf{X}\|_*}_{\text{nonsmooth regularizer } R(\mathbf{X})} \tag{0.1}$$

- Again we can solve it by proximal gradient descent method
- The key is to compute the proximity operator

$$\text{prox}^{\gamma}_{\|\|\|_*}(\mathbf{Y}) = \arg\min_{X\in\mathbb{R}^{m\times n}} \quad \gamma\|\mathbf{X}\|_* + \frac{1}{2}\|\mathbf{X}-\mathbf{Y}\|^2$$

- If this is easily done, then the proximal gradient algorithm is easy

$$\mathbf{X}^{r+1} = \text{prox}^{\lambda/L}_{\|\|\|_*}\left(\mathbf{X}^r - \frac{1}{L}\nabla g(\mathbf{X}^r)\right)$$

The Proximity Operator For Nuclear Norm

- Let's consider solving the problem

$$\min_{X \in \mathbb{R}^{m \times n}} \quad \gamma \|\mathbf{X}\|_* + \frac{1}{2}\|\mathbf{X} - \mathbf{Y}\|^2$$

- The optimal solution is very simple
  1. First compute SVD of $\mathbf{Y}$: $\mathbf{Y} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$
  2. Then define a diagonal matrix $\mathbf{\Sigma}(\gamma)$ as

$$\mathbf{\Sigma}(\gamma)_{ii} = \max\left(\mathbf{\Sigma}_{ii} - \gamma, 0\right)$$

  3. Set $\mathbf{X}^* = \mathbf{U}\mathbf{\Sigma}(\gamma)\mathbf{V}^T$
- Perform "soft-thresholding" on the set of singular values!
- Compare with the ℓ_1 minimization problem we seen before (whose solution is rm Soft$(\mathbf{b}, \gamma)$)

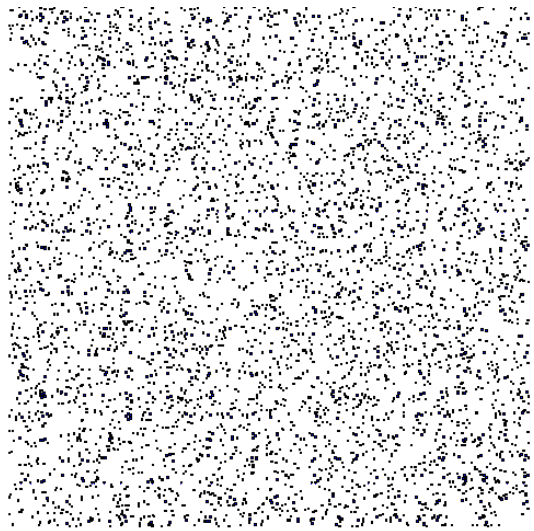
$$\min_{x} \quad \gamma \|\boldsymbol{x}\|_1 + \frac{1}{2}\|\boldsymbol{x} - \mathbf{b}\|^2$$

An Example

- Consider the collaborative filtering problem

$$\min \sum_{(ij)\in\mathcal{I}} (X_{ij} - M_{ij})^2 + \lambda \|\mathbf{X}\|_* \tag{0.2}$$

- **Requirement** the observed entries to be approximately identified $X_{ij} \approx M_{ij}$, the rest is free
- Nuclear norm added to enforce low rank in $\mathbf{X}$
- Specifically let $m = n = 500$, $|\mathcal{I}| = 5000$



Convergence

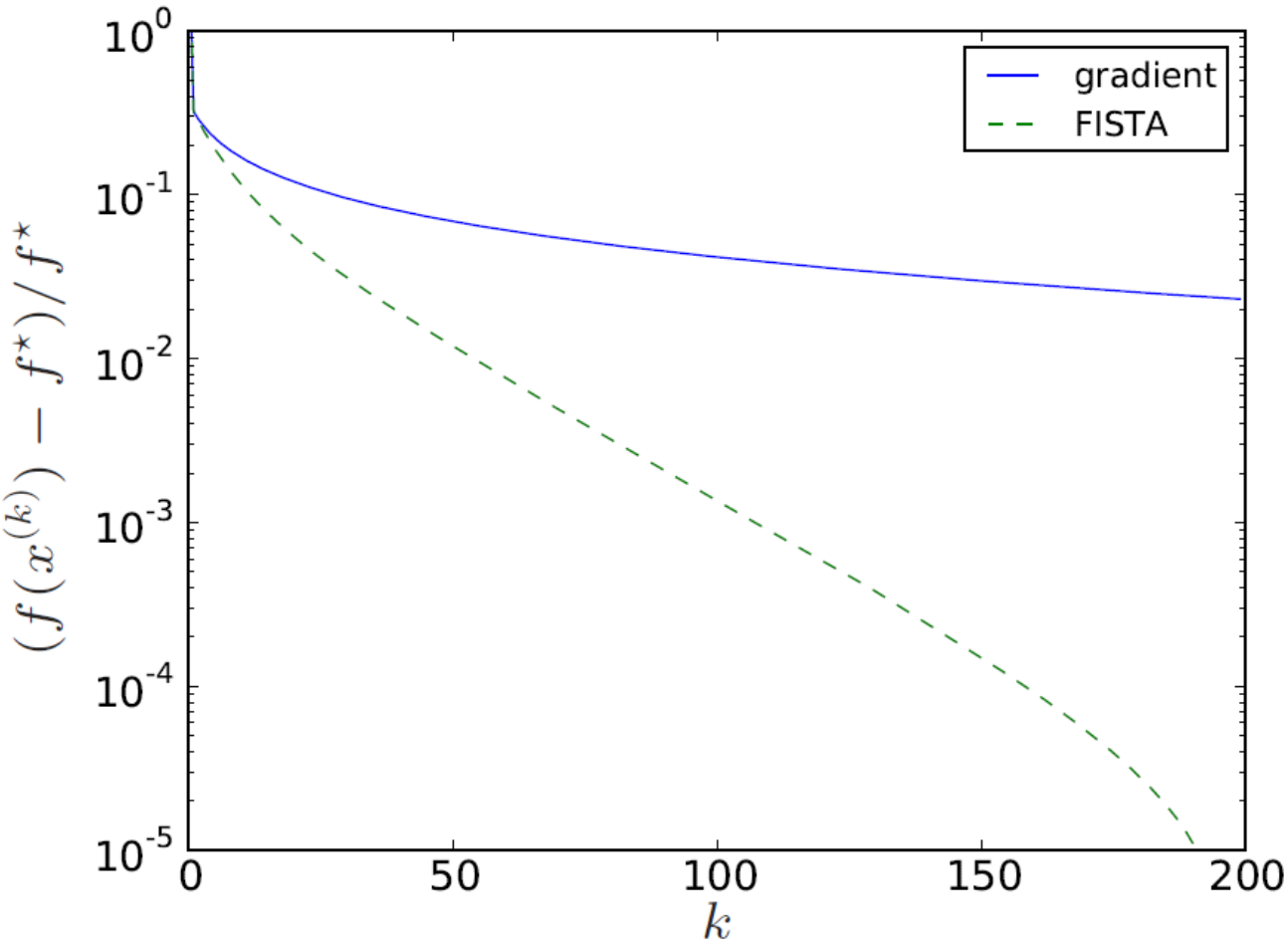


Figure 0.1: Convergence Speed (Proximal Gradient vs. its accelerated version) [Ghaoui10]

The Obtained Rank

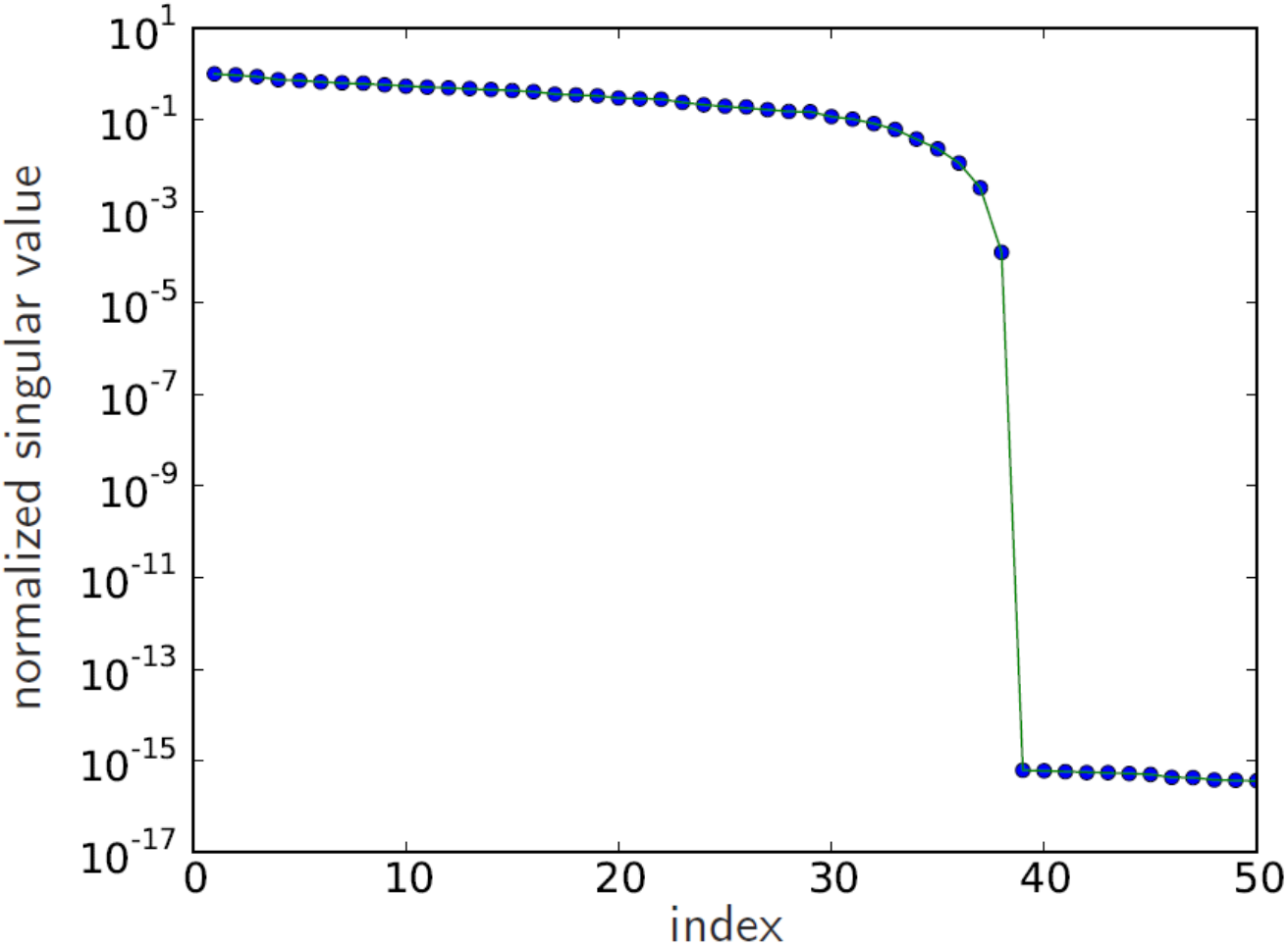


Figure 0.2: Optimal $\mathbf{X}$ has rank 38; relative error in specified entries is 9% [Ghaoui10]